Which of the following muscle types could be assessed for damage by examining the disruption of light and dark banding patterns using light microscopy?

I. Skeletal muscle

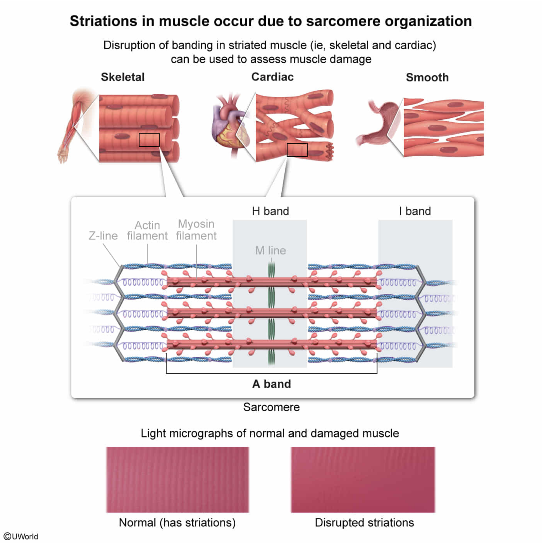
II. Cardiac muscle

III. Smooth muscle

- A. I only [18%]
- B. III only [2%]
- C. I and II only [71%]
- D. I and III only [7%]

Correct
72% answered correctly

Explanation:



Muscles produce force or movement, and can be classified into three types:

- **Cardiac muscle** is found in the walls of the heart, and is specialized for the continuous, rhythmic pumping of blood.
- **Skeletal muscle** is found throughout the body, typically in muscles attached to bones, and exhibits a wide diversity of specialization to allow movements requiring varying degrees of force, velocity, and endurance.
- **Smooth muscle** is found in internal organs and tissues and also exhibits a wide range of specialization to allow periodic or sustained contractions, typically to squeeze or compress a tubular structure (eg, blood vessel, gastrointestinal tract, uterus).

All muscle types produce force through the interaction of actin ("thin") and myosin ("thick") filaments, but these filaments are not organized in the same way in all muscle types. In **cardiac** and **skeletal muscle**, the **arrangement of actin** and **myosin filaments** within sarcomeres results in regular, aligned regions in which the actin and myosin filaments overlap next to regions without overlap.

When examined under a light microscope, the arrangement of actin and myosin filaments in skeletal and cardiac muscle **forms regular repeating patterns** of lighter and darker regions **called striations**. For this reason, cardiac and skeletal muscles are called **striated muscles** in contrast to smooth muscle, which lacks striations.

The question asks which muscle types could be assessed for damage by examining the disruption of light and dark banding patterns using light microscopy. Because striated muscles normally exhibit such patterns due to the regular arrangement of their myosin and actin filaments, **disruption of** these **banding patterns** are **one means** of **assessing damage to cardiac** or **skeletal muscle (Numbers I and II)**.

(Number III) Because actin and myosin filaments are not organized in a regular arrangement, smooth muscle lacks striations. Therefore, using light microscopy to examine the disruption of light and dark banding patterns would *not* be an appropriate way to assess damage in smooth muscle.

Educational objective:
In skeletal and cardiac muscle, the arrangement of actin and myosin filaments within sarcomeres results in regular, aligned regions in which the filaments overlap next to regions in which overlap does not occur. This arrangement results in a regular banding pattern (ie, striations) when skeletal or cardiac muscle is examined under a microscope, and for this reason cardiac and skeletal muscles are called striated muscles.

Time spent: 0 sec	**QID:** 404157
Subject: Biology	**System:** Musculoskeletal System